

# Functions

A function is a piece of code that performs a specific task. It bundles up the code, gives it a name, and can be used any time by “calling” it. A function can be used to avoid entering the same lines of code more than once.

## SEE ALSO

Silly **132–133** ›  
sentences

Variables and **138–139** ›  
functions

## Useful functions

Python contains lots of useful functions for performing certain tasks. When a function is called, Python retrieves the code for that function and then runs it. When the function is finished, the program returns to the line of code that called it and runs the next command.

### print()

#### △ “print()” function

This function lets the program send output to the user by printing instructions or results on the screen.

### input()

#### △ “input()” function

This function is the opposite of the “print()” function. It lets the user give instructions or data to the program by typing them in.

### randint()

#### △ “randint()” function

This function gives a random number (like throwing a dice). It can be used to add an element of chance to programs.

## Making and calling functions

The functions that come with Python aren’t the only ones that can be used. To make a new function, collect the code you want to use in a special “wrapper” and give it a name. This name allows the function to be called whenever it is needed.



### 1 Define a function

The definition of a function will always have the keyword “def” and the function’s name at the beginning of the code.

```
def greeting():  
    print('Hello!')
```

This is the code  
within the function

A colon marks the end  
of the function’s name  
and the start of the  
code it contains

### 2 Call the function

Typing the function name followed by parentheses into the shell window calls the function and shows the output.

```
>>> greeting()  
Hello!
```

The “greeting”  
function is called  
and the output  
is displayed

Parentheses  
show that this is  
a function call,  
not a variable